



Name: \_\_\_\_\_

Date: \_\_\_\_\_

## SYLLOGISMS

Solve the following problems. Write your answer on the line.

Score: \_\_\_\_\_ / 20

Grade 8 | Reasoning | Sheet 3

|                                                                                    |                                                                    |                                                                                          |                                                                               |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 1) All A are B. All B are C. Therefore: _____                                      | 2) All dogs are animals. All animals are dogs. Therefore: _____    | 3) Some kangaroos are marsupials. All kangaroos are marsupials. Therefore: _____         | 4) No A are B. All B are C. Therefore: _____                                  |
| 5) All roses are flowers. Some flowers fade. Therefore: _____                      | 6) All cats are mammals. Leo is a cat. Therefore: _____            | 7) Some students are smart. Rich is a student. Therefore: _____                          | 8) All metals conduct electricity. Iron is a metal. Therefore: _____          |
| 9) No birds are mammals. Eagle is a bird. Therefore: _____                         | 10) All A are B. X is not B. X is _____                            | 11) Some men are brave. No cowards are brave. Therefore: _____                           | 12) All squares are rectangles. Some rectangles are squares. Therefore: _____ |
| 13) All teachers are educated. Some educated people are teachers. Therefore: _____ | 14) All A are B. Some C are A. Therefore: _____                    | 15) Some kangaroos are marsupials. All kangaroos are marsupials. Therefore: _____        | 16) All squares are rectangles. Some rectangles are squares. Therefore: _____ |
| 17) Some boys play cricket. Rahul is a boy. Therefore: _____                       | 18) All planets orbit the Sun. Earth is a planet. Therefore: _____ | 19) All mammals are warm-blooded. All warm-blooded animals are mammals. Therefore: _____ | 20) All fruits are sweet. Mango is a fruit. Therefore: _____                  |



Name: \_\_\_\_\_

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## ANSWER KEY

### Syllogisms -- Solutions

Grade 8 | Reasoning | Sheet 3

|                                                                                                     |                                                                                     |                                                                                           |                                                                                                                    |
|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1) All A are B. All B are C. Therefore: <b>C</b>                                                    | 2) All dogs are animals. All animals are living. Therefore: <b>living</b>           | 3) Some cats are dogs. All dogs are cats. Therefore: <b>C</b>                             | 4) All A are B. All B are C. Therefore: <b>C</b>                                                                   |
| 5) All roses are flowers. Some flowers fade. Conclusion: <b>Some roses may fade</b>                 | 6) All cats are mammals. Leo is a cat. Conclusion: <b>a mammal</b>                  | 7) Some students are smart. Ria is a student. Conclusion: <b>possibly smart</b>           | 8) All metals conduct electricity. Iron is a metal. Conclusion: <b>conducts electricity</b>                        |
| 9) No birds are mammals. Eagle is a bird. Conclusion: <b>not a mammal</b>                           | 10) All A are B. X is not B. X is <b>not A</b>                                      | 11) Some men are brave. No cowards are brave. Conclusion: <b>Some men are not cowards</b> | 12) All P are Q. No Q are R. Therefore: <b>not a conclusion</b>                                                    |
| 13) All teachers are educated. Some educated are rich. Conclusion: <b>Some teachers may be rich</b> | 14) All A are B. Some C are A. Conclusion: <b>B</b>                                 | 15) No fish can walk. Tuna is a fish. Conclusion: <b>cannot walk</b>                      | 16) All squares are rectangles. Some rectangles are squares. Conclusion: <b>Some squares may be parallelograms</b> |
| 17) Some boys play cricket. Rahul is a boy. Conclusion: <b>may play cricket</b>                     | 18) All planets orbit the Sun. Earth is a planet. Conclusion: <b>orbits the Sun</b> | 19) All M are N. All N are O. Therefore: <b>O</b>                                         | 20) All mangoes are sweet. Mango is a fruit. Conclusion: <b>sweet</b>                                              |